Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_11

1. Summary of Key Deviations

The analysis of Run_11 reveals a major physical fault characterized by a severe downstream blockage, leading to a massive redistribution of fuel flow.

- **Flow_Bypass (Major):** The most significant anomaly is a massive positive **offset**, with bypass flow reading up to **+75.8%** higher than healthy. This indicates the vast majority of the fuel is being diverted away from the engines.
- **Flow_Engine1-4 (Moderate):** All four engine flows show a major, consistent **negative offset** of approximately **-26% to -28%**. This confirms that fuel is being severely restricted from reaching the combustion chamber.
- **Flow_Pump (Minor):** The pump flow shows a slight positive **offset** of **+8.7%**. This suggests the pump is working slightly harder to compensate for the high back-pressure from the blockage.
- **Pressure Signals (Minor/Moderate):** System pressures (`p_FMU`, `p_Shut`) show a moderate **negative offset**. This is counter-intuitive but suggests a complex interaction where the massive bypass flow is causing localized pressure drops.
- **Flow_PRV & p_Combustion (Nominal):** There are no significant deviations in the PRV flow or combustion pressure, indicating the system is managing the fault without a dangerous over-pressurization or combustion issue.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Fuel Metering Unit (FMU)**	**Severe Partial Blockage / Restricted Opening**	This is the most likely root cause. The combination of **very low engine flow** (`Flow_Engine` is ~-27%), **extremely high bypass flow** (`Flow_Bypass` is +76%), and **slightly elevated pump activity** (`Flow_Pump` is +9%) creates a classic signature of a major downstream restriction. The FMU is the primary component capable of causing such a uniform blockage across all four engine lines.

2	**Injector Nozzles (All Four)**	**Simultaneous Partial Clogging**	It's possible that all four injector nozzles became partially clogged at the same time. This would produce the same symptoms as a blocked FMU. However, a single point of failure in the upstream FMU is generally more probable than four simultaneous, identical failures in the nozzles.
3	**Shut-off Valve**	**Partial Blockage**	A partial blockage in the shut-off valve, which is downstream of the FMU, could also cause a system-wide restriction. It is another plausible location for the blockage.
4	**Bypass Valve**	**Incorrect Calibration / Sticking**	The bypass valve is opening very wide to divert the flow that cannot pass the blockage. This is a *symptom* of the blockage, not the root cause. The valve is reacting correctly to the high back-pressure.
5	**System Contamination**	**Debris causing blockage**	This is the likely underlying *cause* of the blockage itself (e.g., a piece of debris lodging in the FMU). The primary diagnostic task is to identify the *location* of the blockage, which is clearly downstream of the bypass tee and before the engine nozzles.

3. Most Probable Hypothesis

The most probable root cause is a **Severe Partial Blockage within the Fuel Metering Unit (FMU).**

Reasoning: The data provides a clear, textbook case of a major downstream restriction. The three key symptoms—**severely decreased engine flow, massively increased bypass flow, and increased pump activity**—all

point to a single conclusion: something is preventing the majority of the fuel from getting to the combustion chamber. The FMU is the most logical single point of failure that would affect all four engine nozzles simultaneously and uniformly. The system is correctly detecting the resulting high back-pressure and diverting almost all the pump's output through the bypass loop, as shown by the +76% bypass flow reading.